Introduction to Probability and Statistics

(EE 354 / CE 361)

Random Variables and Discrete Random Variables

Today's Agenda

- Random Variables
- Discrete Random Variables

Recall

- Problems (Counting)

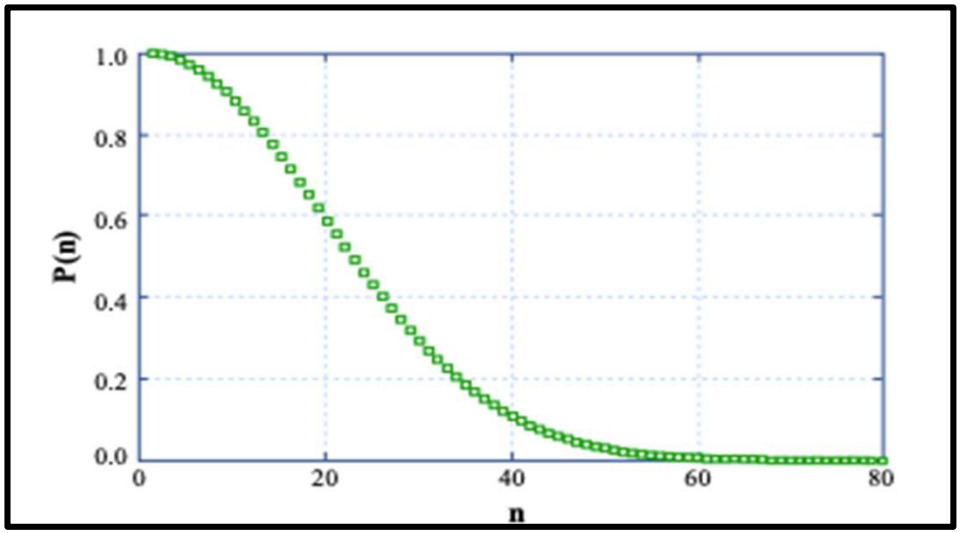
The Birthday Problem

Given there are n Students in a class, find the P(no 2 birthdays coincide)

1 < n < 365

Ω = all possible sequences

| Ω | = 365 ^ n



S = 365 * 364 * … * (365-n+1) = $\frac{365!}{(365-n)!}$

P (no 2 birthdays coincide) = $\frac{365!}{(365-n)!}\frac{1}{(365)^n}$

Practice Problem:

- If I flip a coin 20 times, I get a sequence of Heads (H) and tails (T). How many different sequences have at least three heads?

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Solution: The total number of sequences that are possible here is 2^{20}. We want number of sequences that have at least 3 heads, so we can find out the number of sequences that have no heads, 1 heads, and 2 heads and subtract it from total number of sequences.

$$\text{Number of sequences with no heads} = 1$$

$$\text{Number of sequences with 1 heads} = \binom{20}{1}$$

$$\text{Number of sequences with 2 heads} = \binom{20}{2}$$

Therefore,

$$\text{Number of sequences with at least 3 heads} = 2^{20} - 1 - \binom{20}{1} - \binom{20}{2}$$

We can also approach it by finding the number of sequence with 3 heads, 4 heads, all the way to 20 heads and summing them up, i.e.

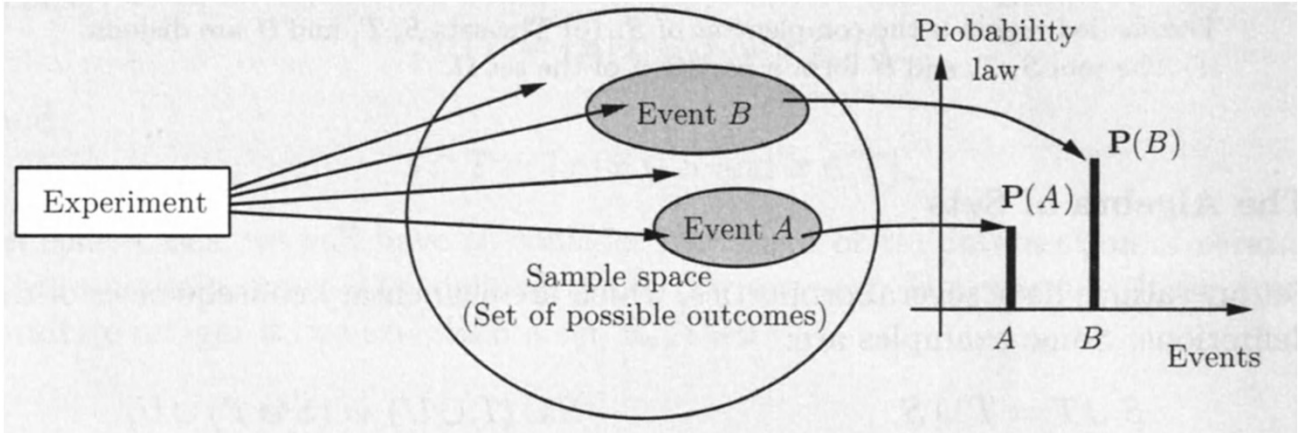
$$\text{Number of sequences with at least 3 heads} = \sum_{k=3}^{20} \binom{20}{k}$$

Probabilistic Model

Every probabilistic model involves an underlying process, called the **Experiment**, that will produce exactly one out of several possible outcomes.

- Sample Space (Ω)
  - The set of all possible outcomes of an experiment

- Probability Law
  - Assigns to a set A of possible outcomes a non-negative number $\mathbf{P}(A)$ that encodes our belief about the collective "likelihood" of the elements of A.
  - This probability law must satisfy certain properties, known as "Probability Axioms."



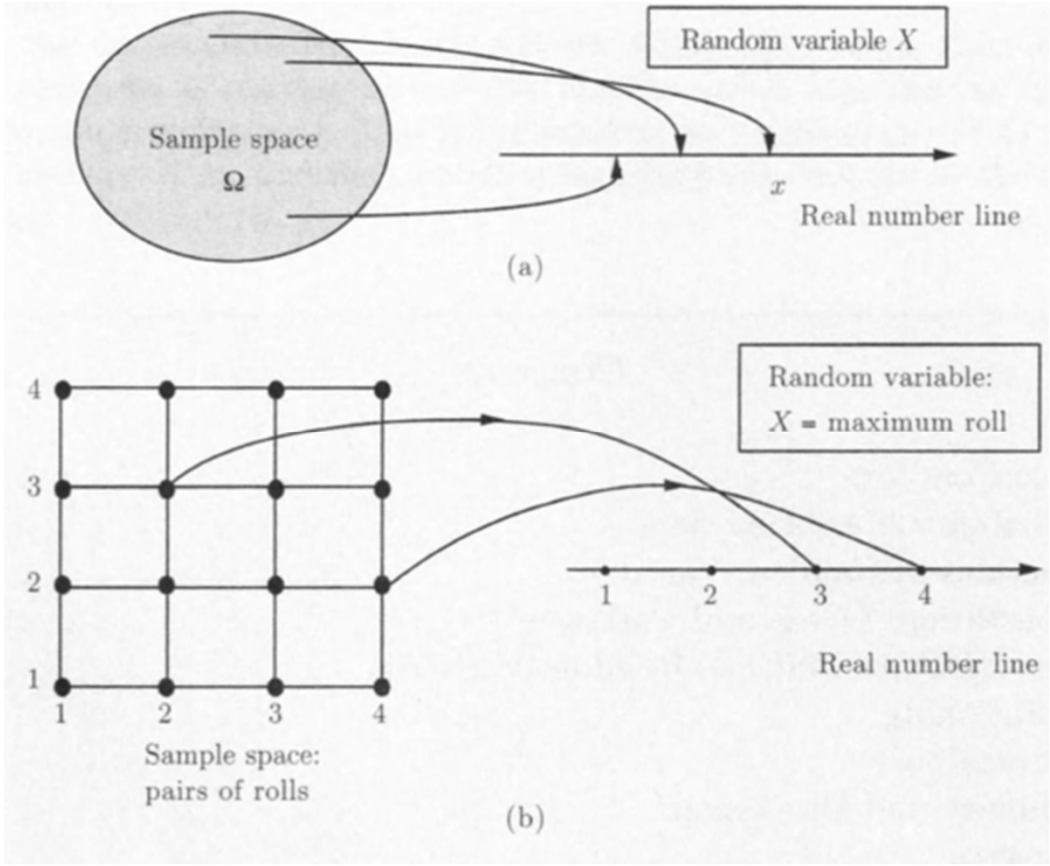
Random Variables: The Need

Random Variables: The Idea

- In a probabilistic model, outcomes of an experiment
  - May be numerical e.g. rolling a 4-sided die (1, 2, 3, 4)
  - May be non-numerical e.g. Flipping a coin (Head or Tail)

- Random Variable
  - Given an experiment and the corresponding set of possible outcomes (the sample space), a random variable assigns a particular real number with each possible outcome.

- Examples
  - Number of heads in a sequence of 5 tosses of a coin
  - For two rolls of die, the maximum of two rolls
  - GPA of a randomly selected student in the class


Random variable X
Sample space
Ω
x
Real number line
(a)
Random variable:
X = maximum roll
4
3
2
1
1
2
3
4
Real number line
Sample space:
pairs of rolls
(b)

Random Variables: The Formalization

- Random Variable
  - Mathematically, a random variable is a function from the sample space to real numbers:

$$X: \Omega \rightarrow \mathbb{R}$$

- Important Distinction:
  - X – random variable ("procedure" / "subroutine")
  - x – numerical value $(x \in \mathbb{R})$

Random Variables: The Formalization

- Random Variable
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- Discrete Random Variables
  - A random variable is called **discrete** if its range (the set of values that it can take) is either finite or countably infinite.

- Continuous Random Variables
  - A random variable is called **continuous** if its range (the set of values it can take) is uncountably infinite.

Multiple Random Variables on the same Sample Space

- We can have multiple random variables defined on the same sample space

- Example
  - Weight of a randomly selected student in the class
  - Height of a randomly selected student in the class

Functions of Random Variables

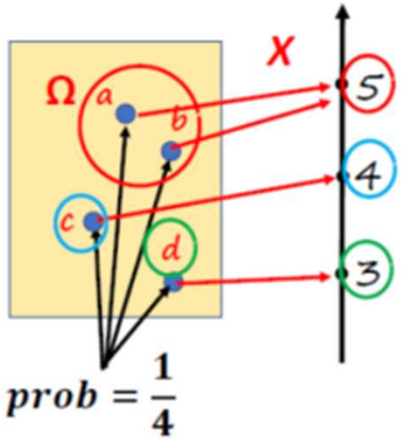
- The function of one or several random variables is also a random variable.

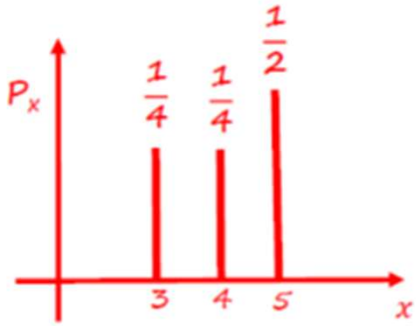
Probability Mass Function (PMF) of a Discrete Random Variable

- How to characterize/describe a random variable
  - The most important way to characterize a random variable is through the probabilities of the values it can take.
  - For a discrete random variable X, this information is captured by its **Probability Mass Function (PMF)**, denoted by: p_X

In particular, if x is any possible value of X, the **probability mass** of x, denoted $p_X(x)$, is the probability of the event $\{X = x\}$ consisting of all outcomes that give rise to a value of X equal to x:

$$p_X(x) = \mathbf{P}(\{X = x\}) = \mathbf{P}(\{w \in \Omega \text{ s.t. } X(w) = x\})$$



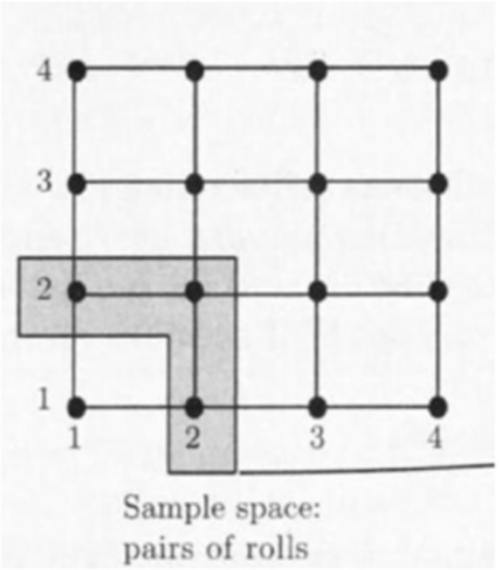


Calculation of the PMF of a Random Variable

For each possible value x of X:

1. Collect all the possible outcomes that give rise to the event $\{X = x\}$.
2. Add their probabilities to obtain $p_X(x)$.

- Example
  - Two rolls of a fair tetrahedral die
  - Random Variable: X = Maximum of two rolls



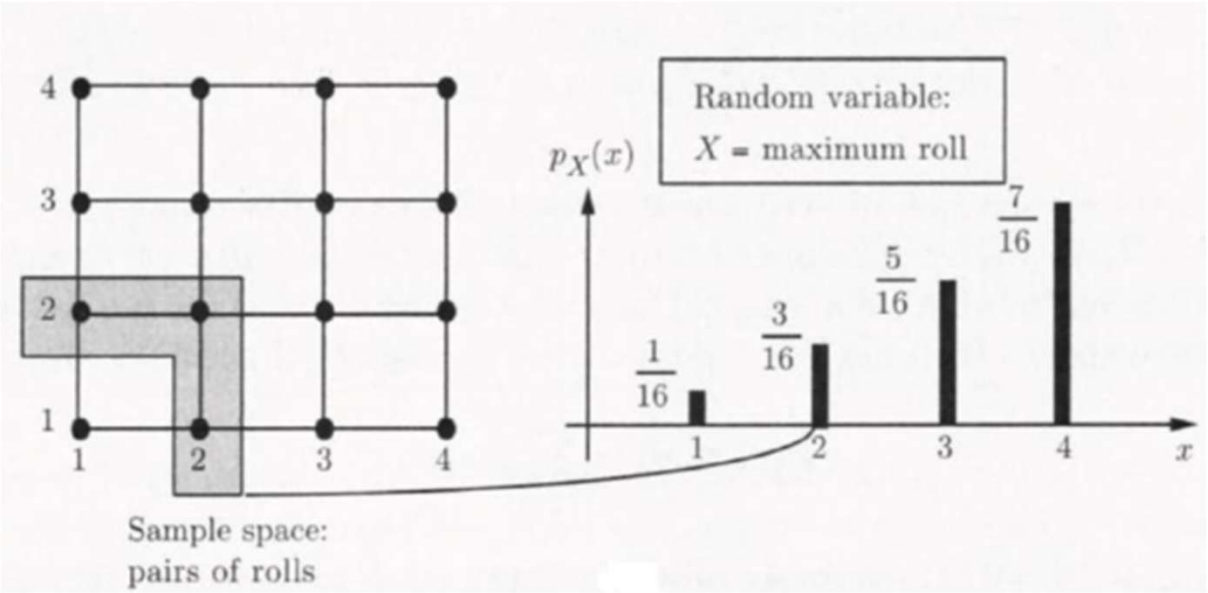
Sample space:
pairs of rolls

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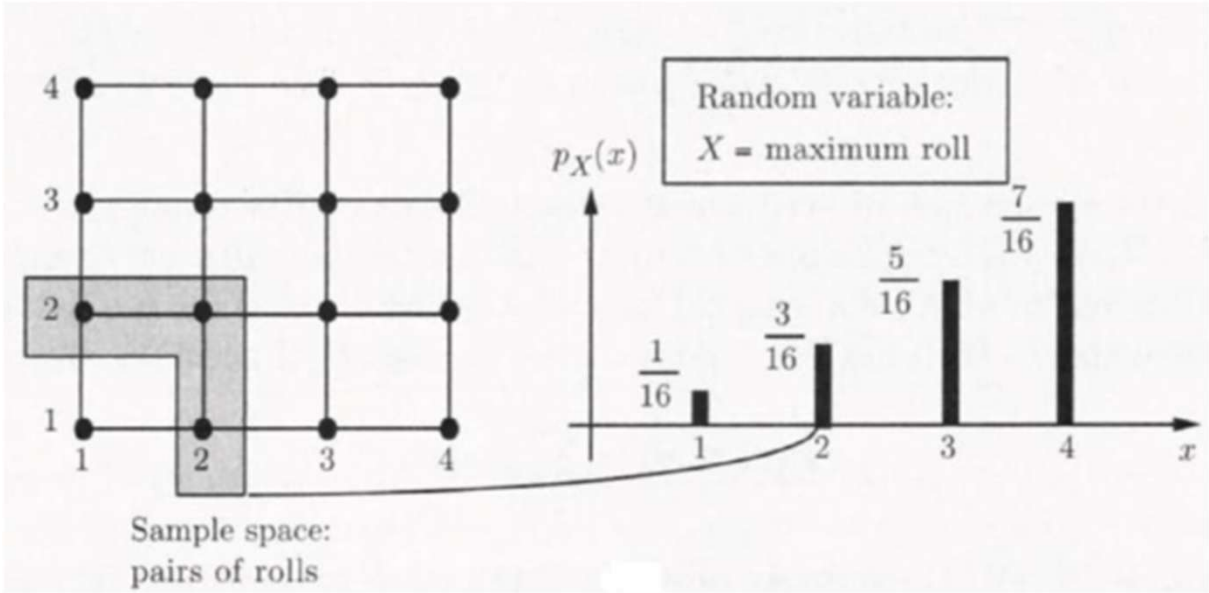


Properties of the PMF of a Random Variable

- Properties:

$$p_X(x) \geq 0 \qquad \sum_x p_X(x) = 1$$

- Example
  - Two rolls of a fair tetrahedral die
  - Random Variable: X = Maximum of two rolls



Axioms:

a) $\mathbf{P}(A) \geq 0$

b) $\mathbf{P}(\Omega) = 1$

For disjoint events

c) $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B)$